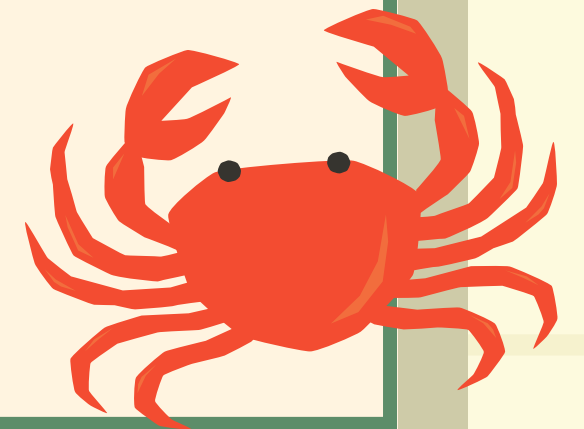
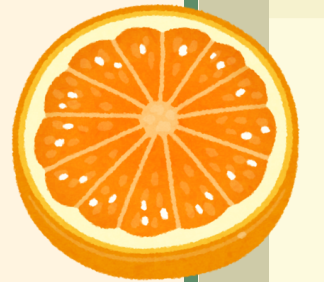
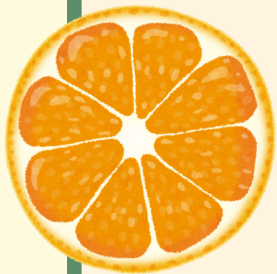


FOOD-101 IMAGE CLASSIFICATION

CV PROJECT

EFFICIENTNETB3 . RESNET101 .
DENSENET201

By Nashra & Kashaf



INTRODUCTION

Dataset: Food-101

- 101 food categories
- 75,750 training images and 25,250 testing images
- Provided train/test split
- Images are real-world, high-variance, non-studio quality
- Used original authors' metadata to build clean directory structure



DEEP LEARNING MODELS

We trained three transfer learning models:

1. EfficientNetB3

- Best parameter efficiency
- Prefers 300×300 but trained at 224×224 for speed

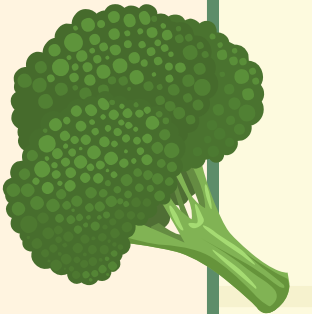
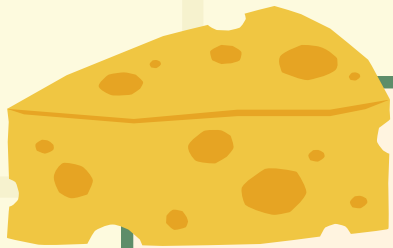
2. ResNet101

- Very deep residual architecture
- Helps avoid vanishing gradients

3. DenseNet201

- Dense connections → strong feature reuse
- Efficient and stable training

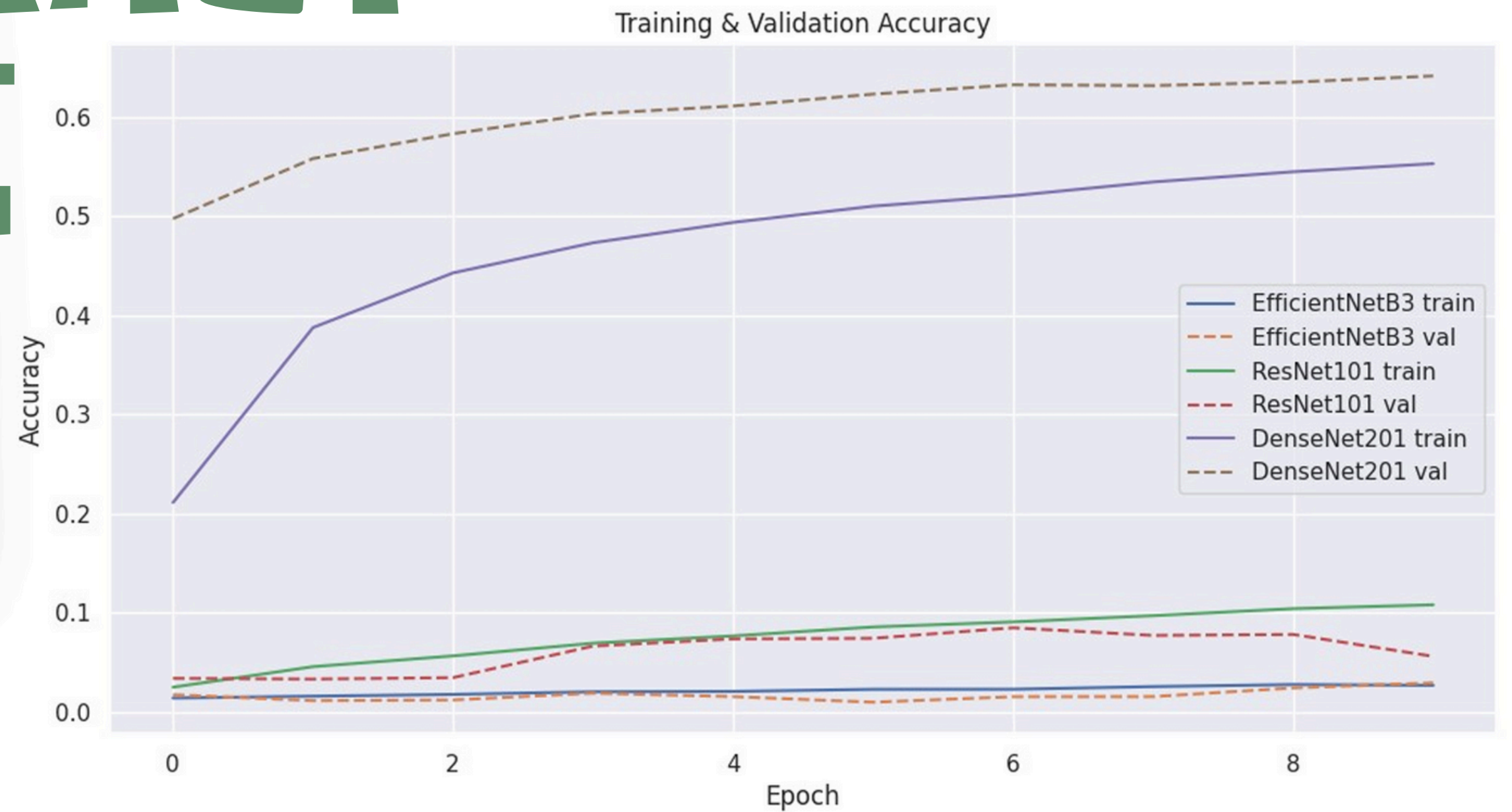




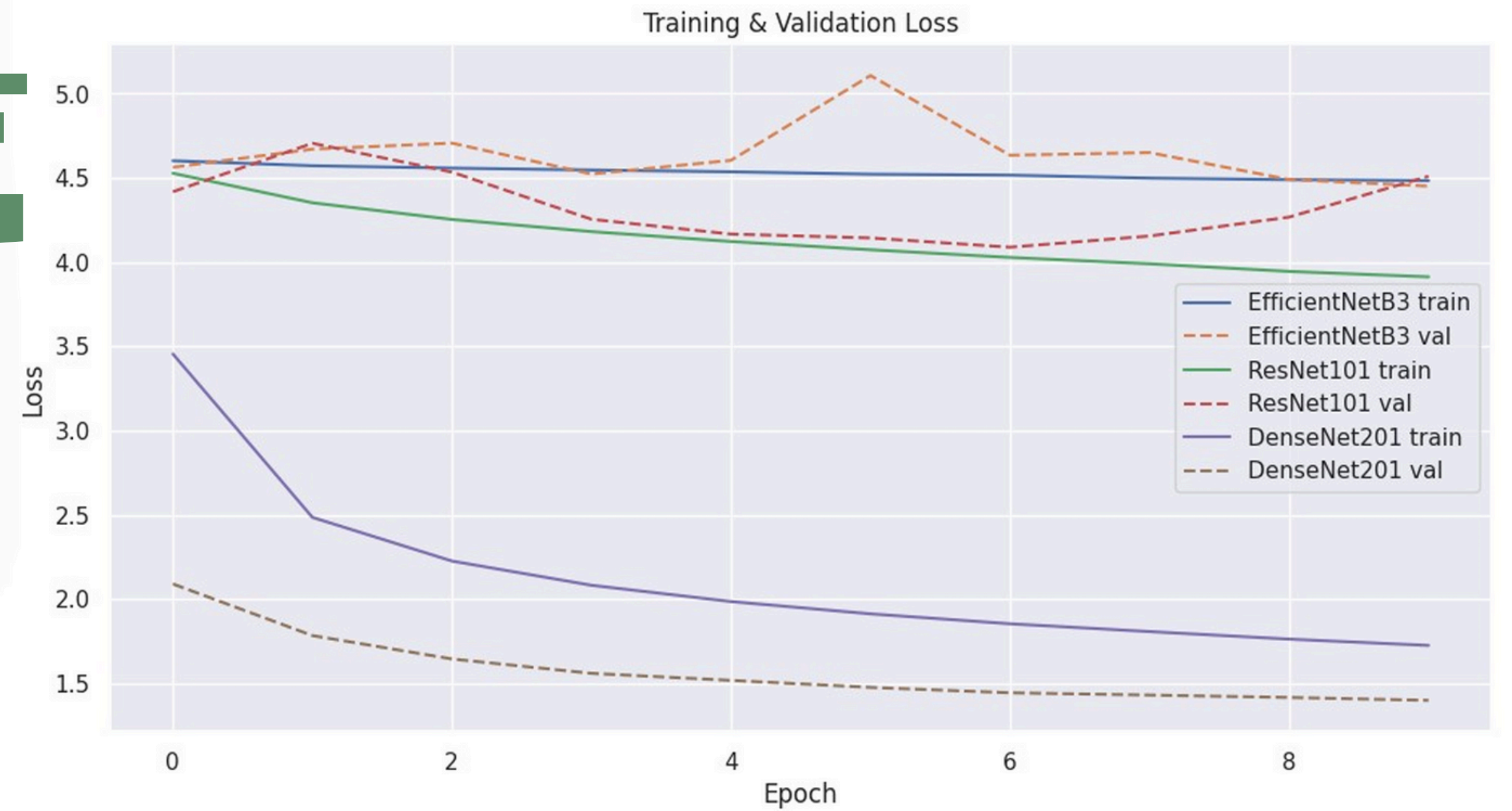
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Loss	Macro Precision	Macro Recall	Macro F1
EfficientNetB3	0.033505	4.411595	0.018741	0.033386	0.012116
ResNet101	0.097703	3.989144	0.093745	0.097545	0.077837
DenseNet201	0.698099	1.118766	0.700891	0.698178	0.693492

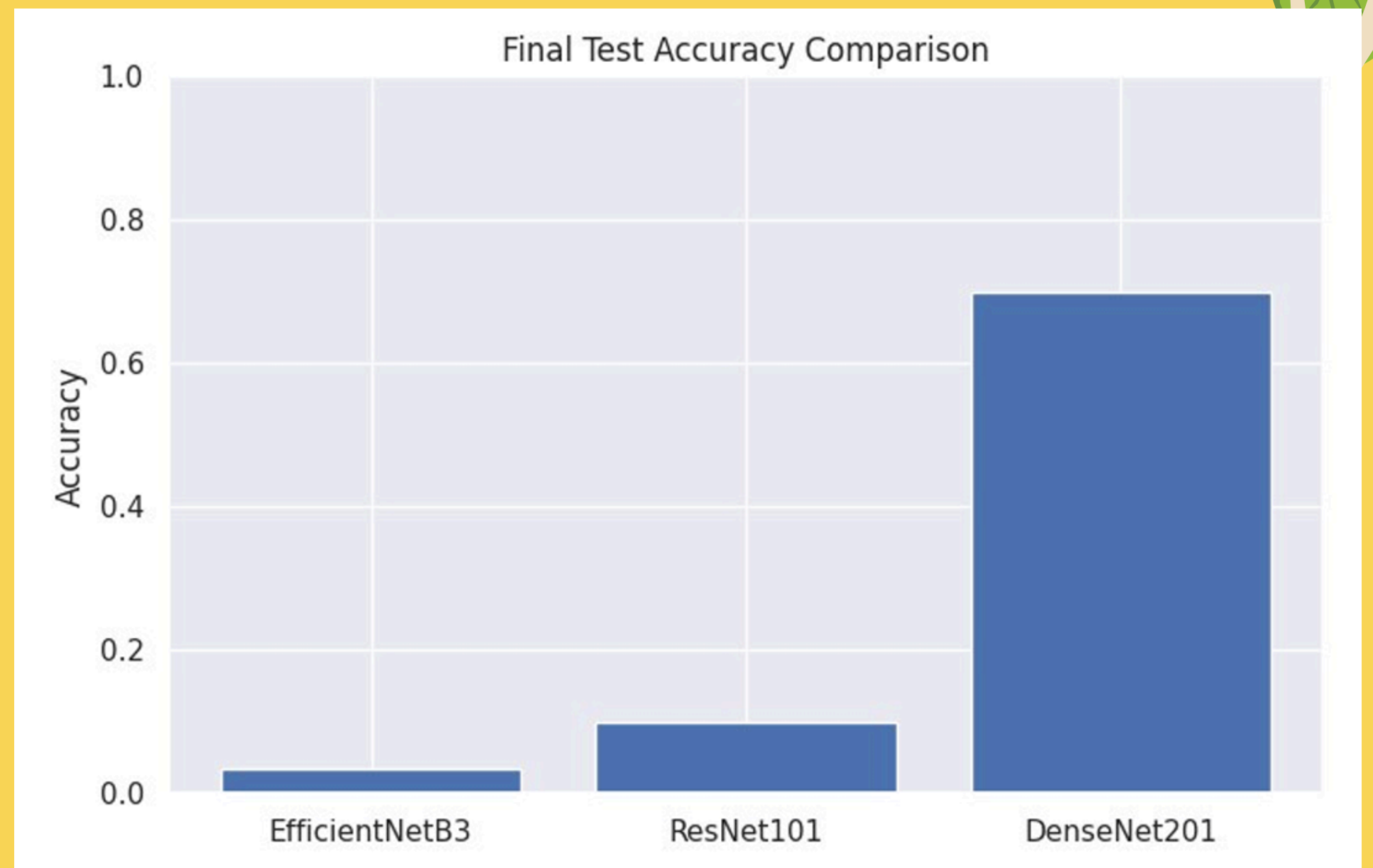
ACCURACY CURVE



LOSS CURVE



FINAL ACCURACY BAR CHART



INFERENCE DEMO



This section shows:

- Three correct predictions
 - Three incorrect predictions
- per model



EFFICIENT NET B3

T: spaghetti_bolognese
P: spaghetti_bolognese



EfficientNetB3 — Correct examples
T: mussels
P: mussels



T: spaghetti_bolognese
P: spaghetti_bolognese



EfficientNetB3 — Incorrect examples
T: poutine
P: edamame

T: carrot_cake
P: miso_soup



T: chicken_curry
P: gnocchi



RES NET 101

T: frozen_yogurt
P: miso_soup



ResNet101 — Incorrect examples
T: breakfast_burrito
P: frozen_yogurt



T: hummus
P: clam_chowder



T: lobster_bisque
P: lobster_bisque



ResNet101 — Correct examples
T: cheese_plate
P: cheese_plate



T: filet_mignon
P: filet_mignon



DENSE NET 201

T: ice_cream
P: ice_cream



DenseNet201 — Correct examples
T: garlic_bread
P: garlic_bread



T: dumplings
P: dumplings



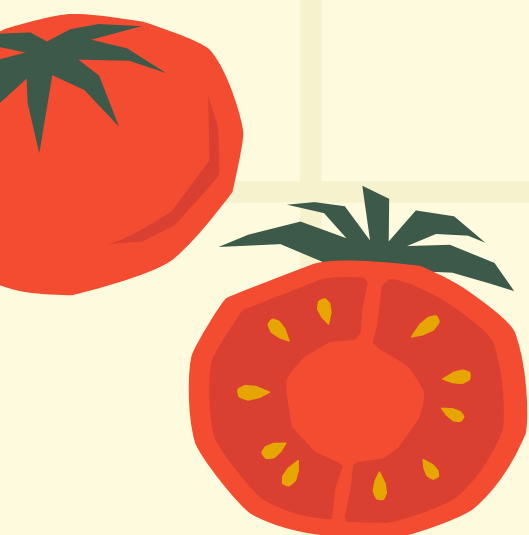
T: waffles
P: onion_rings



DenseNet201 — Incorrect examples
T: ceviche
P: guacamole



T: pulled_pork_sandwich
P: french_fries





LEARNINGS & CONTRIBUTIONS



Technical Learning



- Preprocessing large datasets and creating Keras-friendly directory structures
- Using ImageDataGenerator + augmentation
- Training large CNN architectures on limited GPU time
- Managing model saving, history logs, metrics, and confusion matrices
- Using transfer learning and selective layer unfreezing



LEARNINGS & CONTRIBUTIONS



Personal Contributions

- Dataset organization and preprocessing
- Implemented training pipeline for all models
- Designed comparison framework and metrics extraction
- Developed final comparison notebook
- Conducted inference visualization



CONCLUSION

This project strengthened our understanding of deep CNNs, experiment tracking, and model evaluation.

(also made us cry :_)





THANK YOU

By

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- Nashrah Ahmed (F223156)